



Fig. 4: Google Pixel 2 OLED display



Fig. 5: An OLED TV set

electrons when a potential difference is applied across terminals.

Applications

OLED displays are mainly used in digital devices such as high-end television systems, computer monitors, pocket-size systems such as Android phones, media players, digital cameras, portable gaming consoles and mini-screens. Such types of applications demand high reliability and readability. OLEDs fit in as these consume less power and provide high-quality display. The current OLED technology provides remarkable colour fidelity, high



Fig. 6: Digital camera with OLED screen



Fig. 7: OLED screen on wearable watch



Fig. 8: An OLED desk lamp

efficiency and operation stability.

Products available across different categories are listed below.

Mobile phones. Since year 2016 many mobile phones have been introduced with OLED screen, including Samsung A8+, Gionee S11S and M7 plus, Oppo A79 and R11 plus, OnePlus 5T, Google Pixel 2 and iPhone X.

OLED TV. TV display technology has evolved from LED to OLED and QLED. At present, all companies

are working on OLED displays to make their products smart. OLED TV models include B&O BeoVision Eclipse, Loewe Bild 4, Bild 5 and Bild 9, Toshiba X97, Skyworth W9 Wallpaper OLED and S9E, Panasonic EZ950 and EZ1000, Sony XBR-A1E Bravia and Signage 65EE5PC. Some of the companies like LG have even categorised their TV displays into flat OLED display, curved OLED display and OLED wallpaper.

Digital cameras. Use of OLED displays in digital camera screens or electronic viewfinders is on the rise. The first OLED camera was Kodak LS633 in 2003. Today, there are a host of OLED models available like Canon Power Shot G1 X Mark III, Fujifilm GFX 50S and X-T2, Sony a99 Mark II and HX80, Panasonic DMC-FZ2500 and Lumix DMC-G85, and Nikon DL24-500, DL18-50 and DL24-85.

Wearable devices. Wearable devices include multi-purpose tools that use advanced sensing technologies such as voice command recognition, gesture recognition, and face and fingerprint recognition. OLED technology brings advantages of thinner and curved display form factor to wearable devices. Some of the wearable devices with OLED screens are Xiaomi Mi Band 2, MIO Slice, KIGO Watch, STARVR, Fitbit Alta, I Fit Vue, Mykronoz Zefit2 Pulse and Zefit2, Moto 360 Sport, Asus Zenwatch 2, HTC Vive, Sony PlayStation VR, Razer OSVR, Garmin Vivomart, and Huawei Talk band B1 and Talk band B2.

OLED light panels. OLED technology enables large-area light panels. OLED displays can be used to make flexible, transparent and colour-tunable light panels. OLED panels emit diffused light, which is very close to natural sunlight. OLED light panels from various companies include IKEA Vitsand, SKK Goose-neck, Ascend, Acuity Brands Olessence and Nomi, Workrite Ergonomics Natural OLED, Bernd Unrecht Adjust-S, Pugnale and Nylene Volker OLED Luminaire, and OTI Aerelight OLED. There are light panels available from LG, Philips, Osram and Konica Minolta too. **EFY**